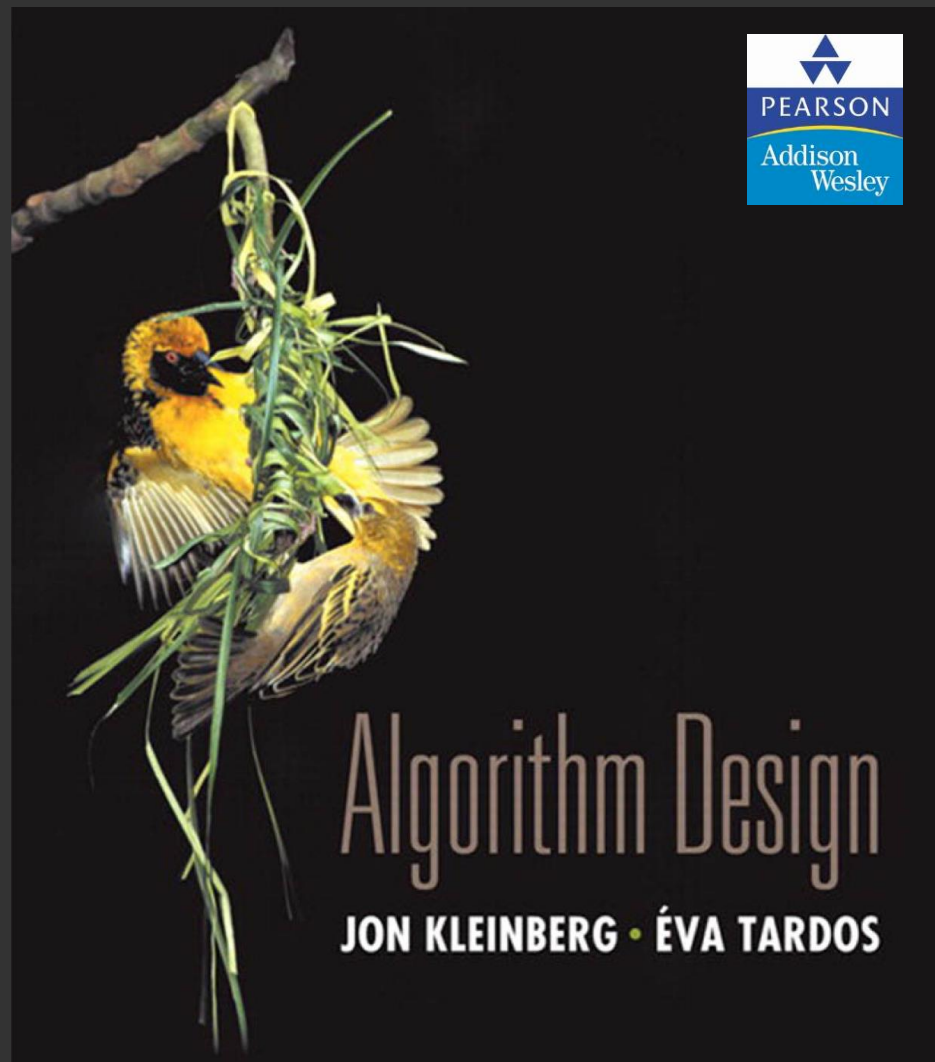




INTRACTABILITY III

- ▶ *greedy independent set algorithm demo*

Lecture slides by Kevin Wayne

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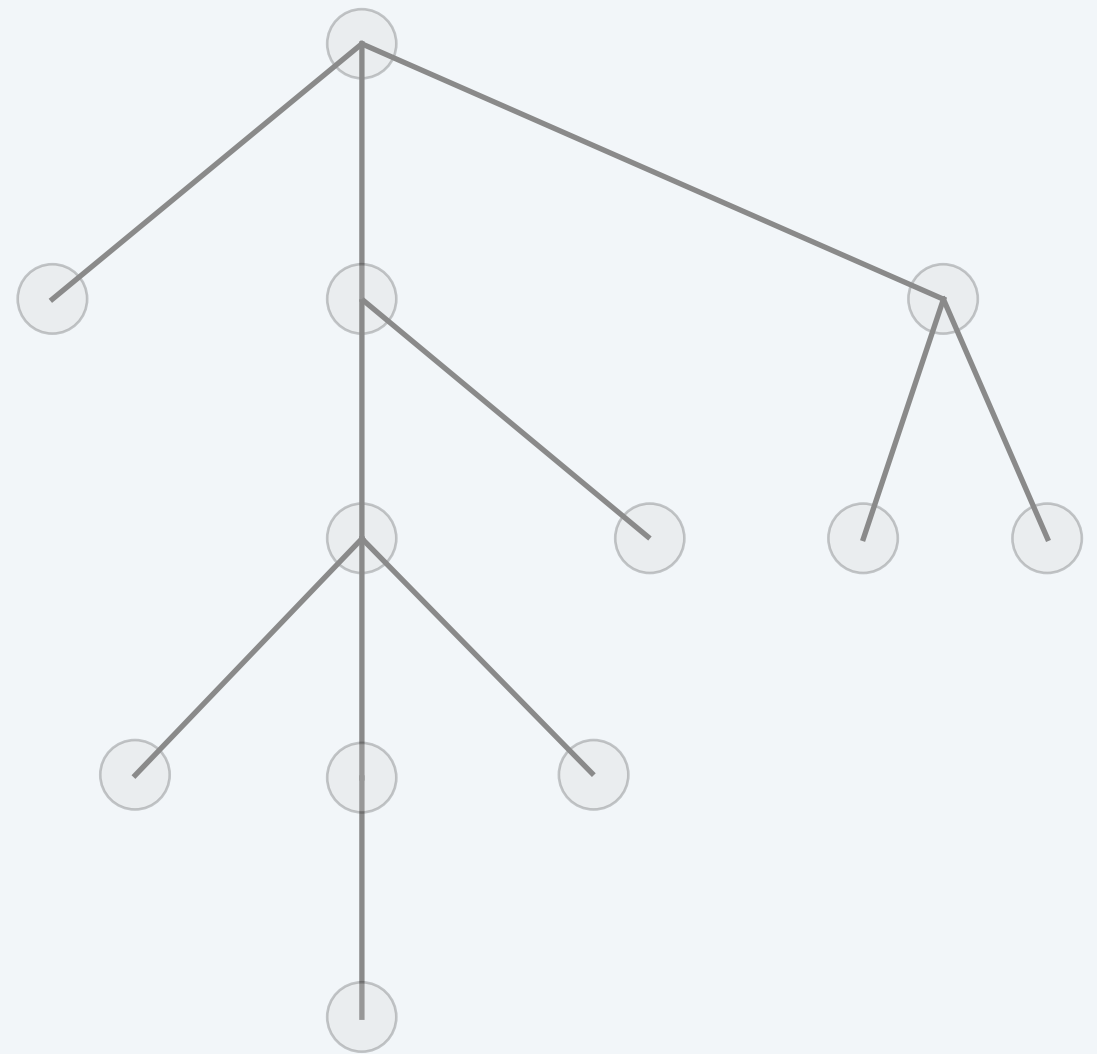
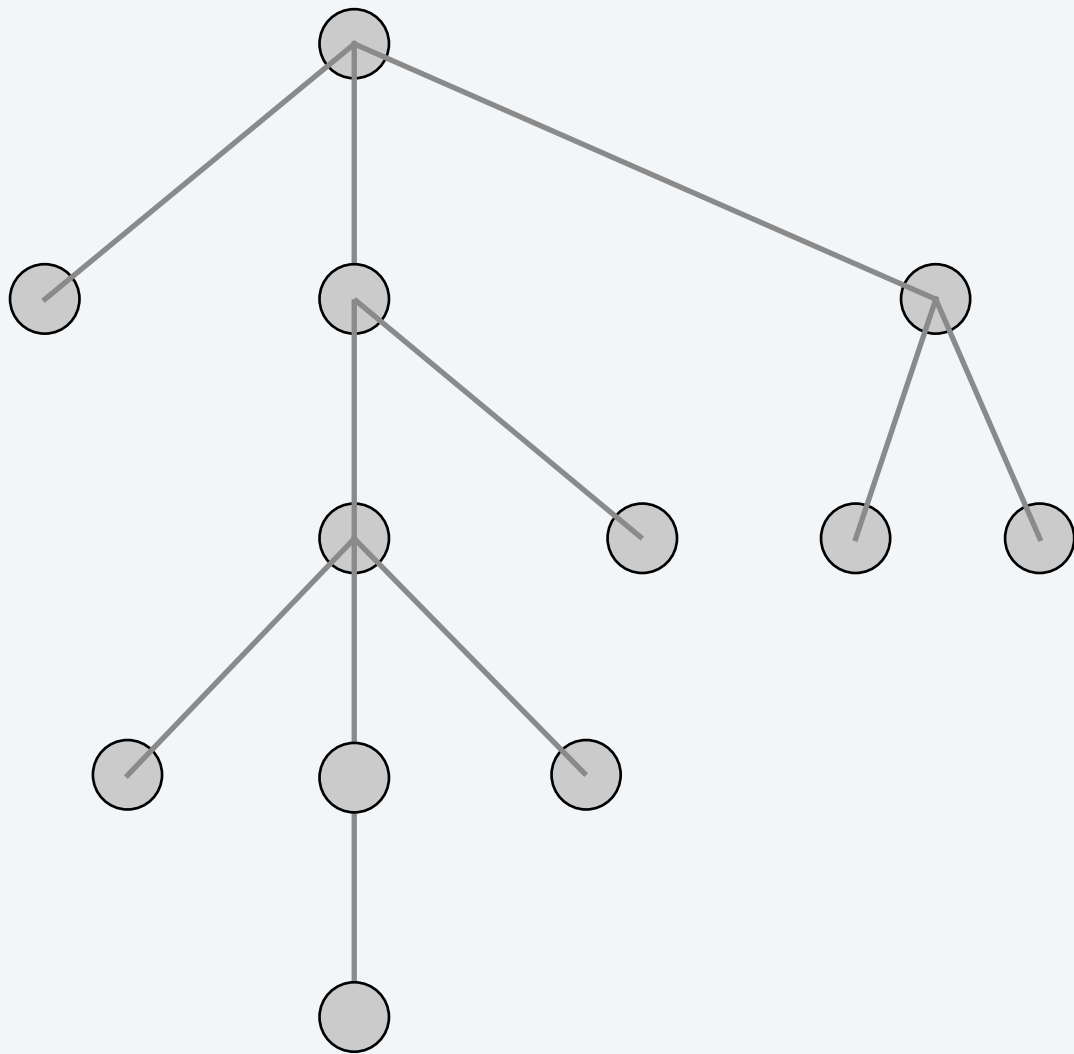
<http://www.cs.princeton.edu/~wayne/kleinberg-tardos>

Abdur Rashid Tushar
Lecturer, CSE, BUET

Greedy independent set algorithm (for trees)

Repeatedly pick a leaf node v .

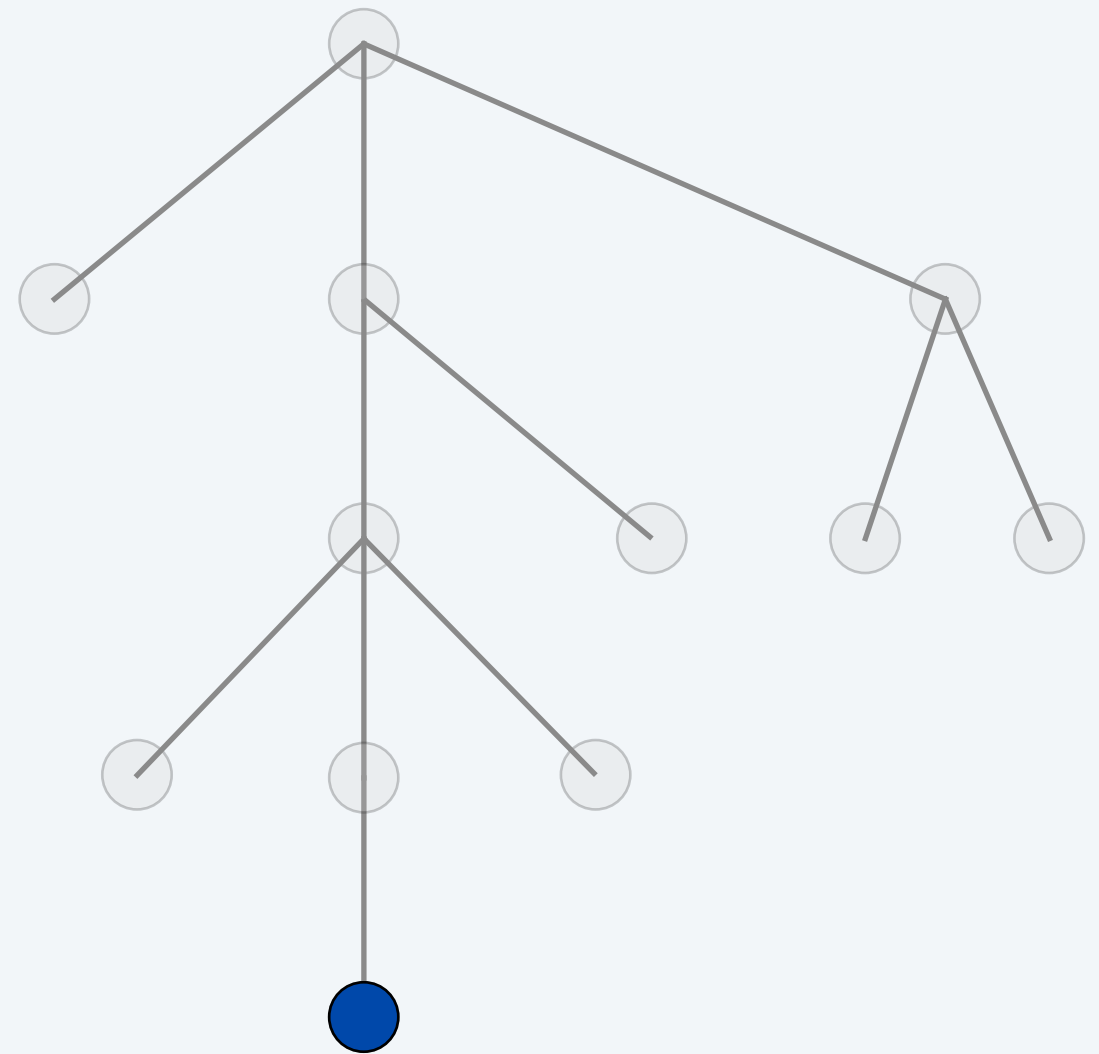
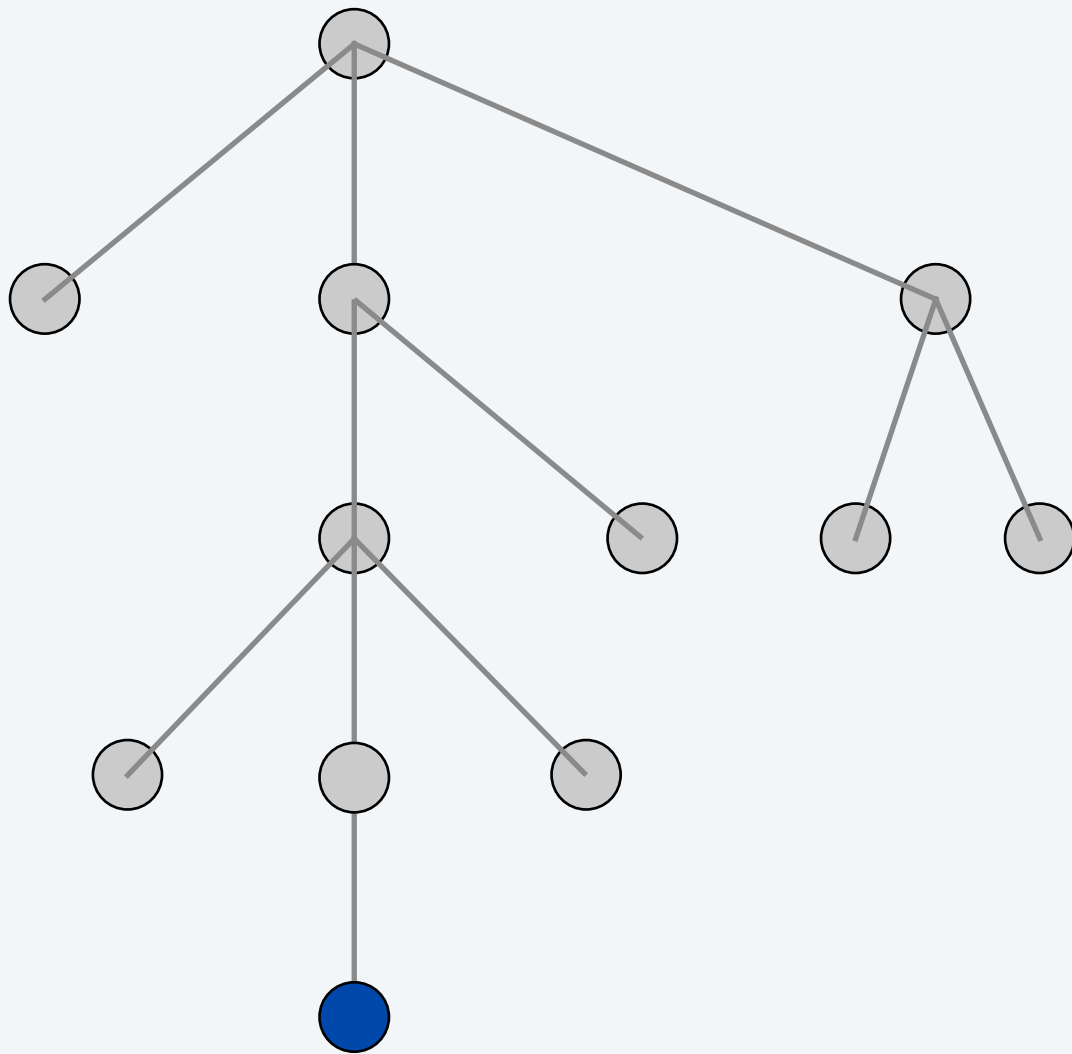
- Let (u, v) denote the incident edge.
- Add both v to the independent set.
- Delete u and v (and all incident edges).



Greedy independent set algorithm (for trees)

Repeatedly pick a leaf node v .

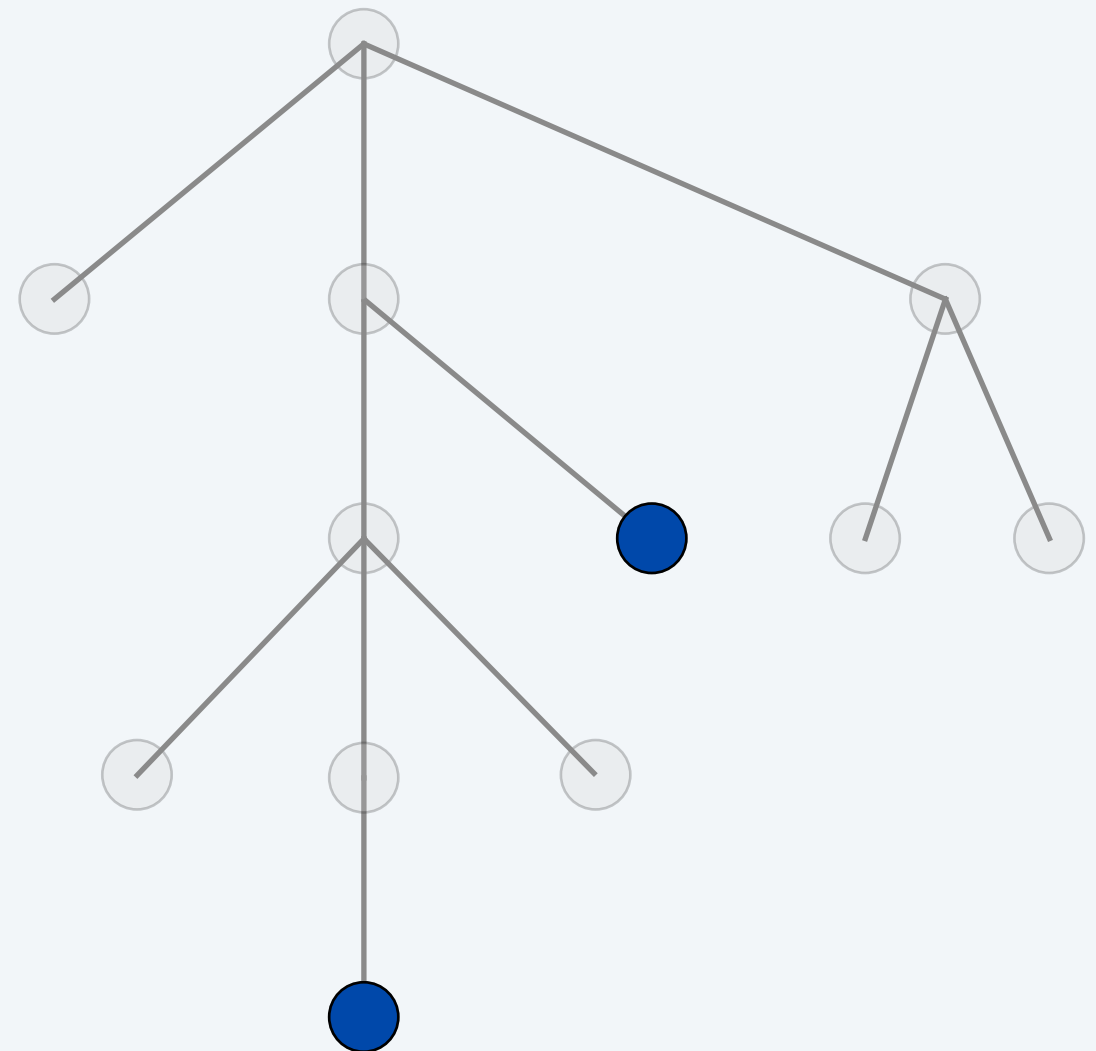
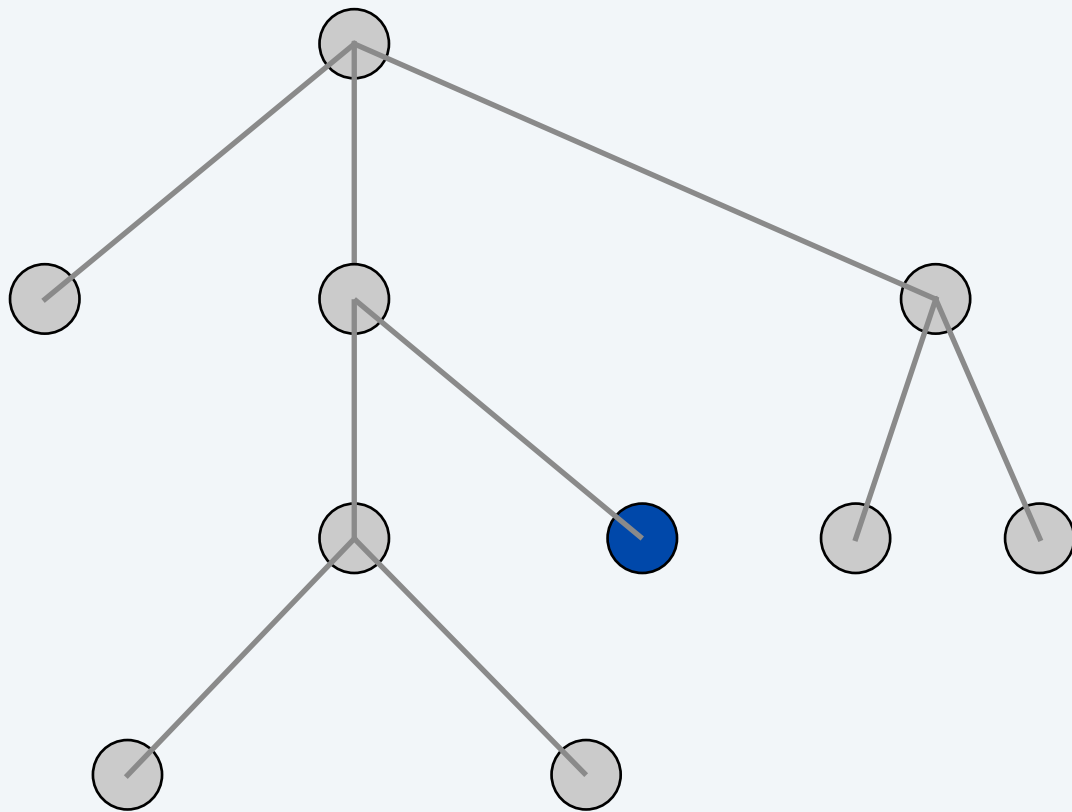
- Let (u, v) denote the incident edge.
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Greedy independent set algorithm (for trees)

Repeatedly pick a leaf node v .

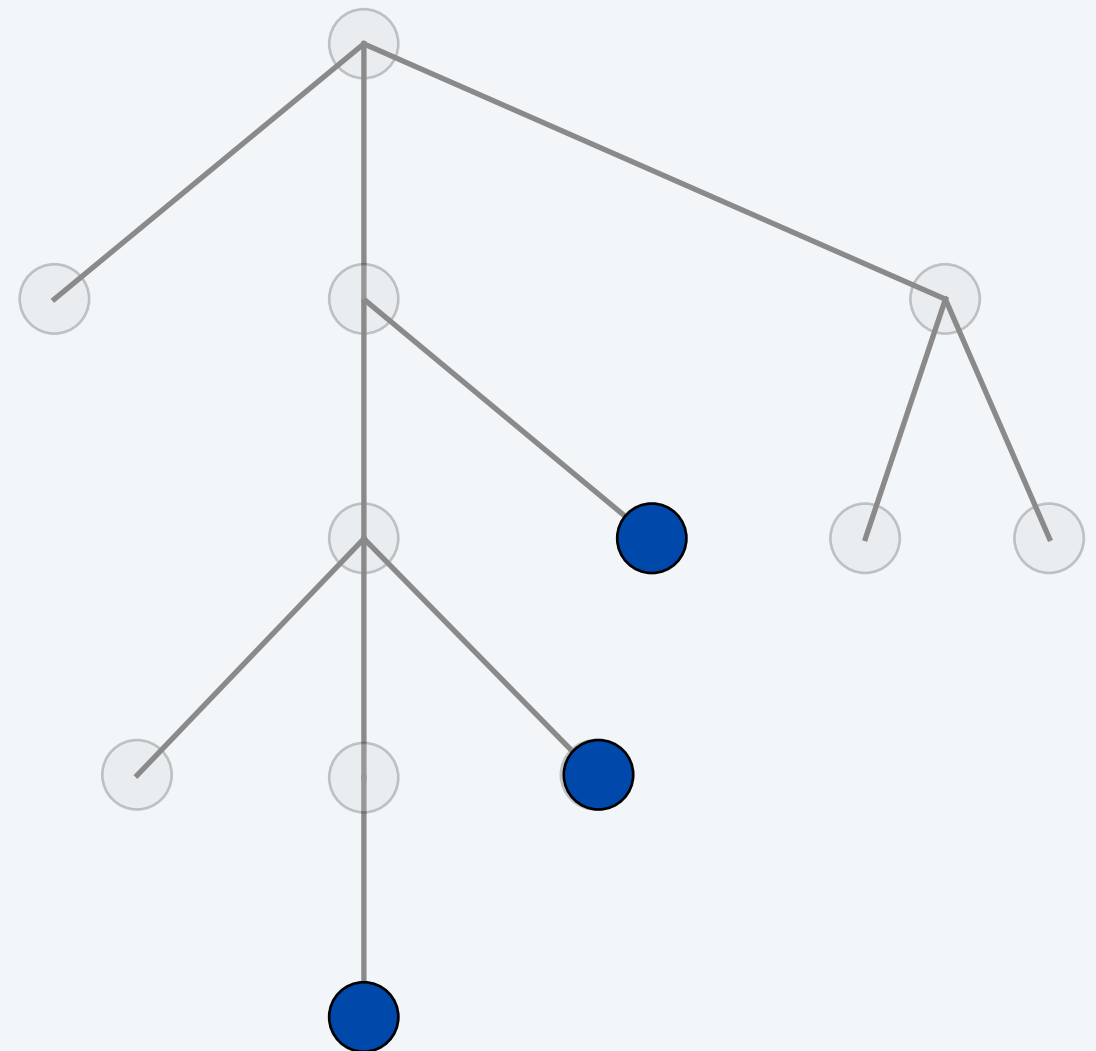
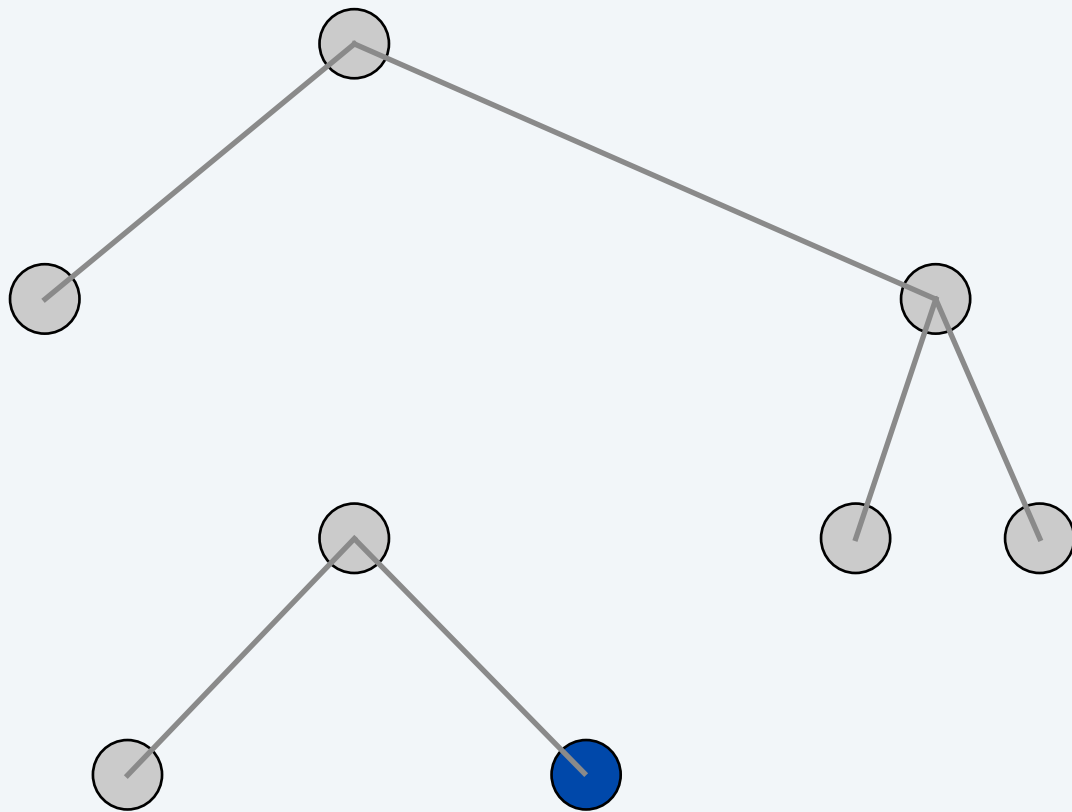
- Let (u, v) denote the incident edge.
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Greedy independent set algorithm (for trees)

Repeatedly pick a leaf node v .

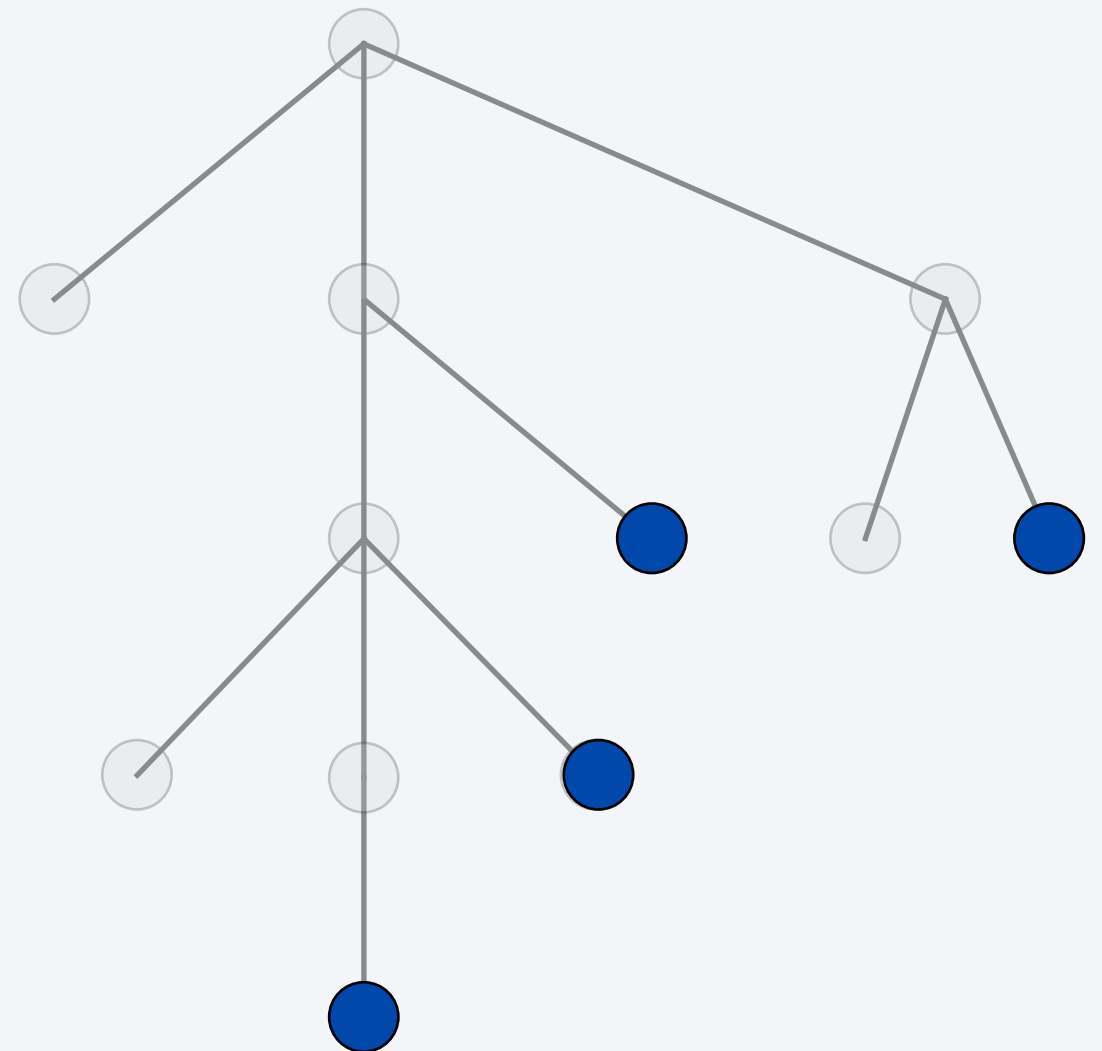
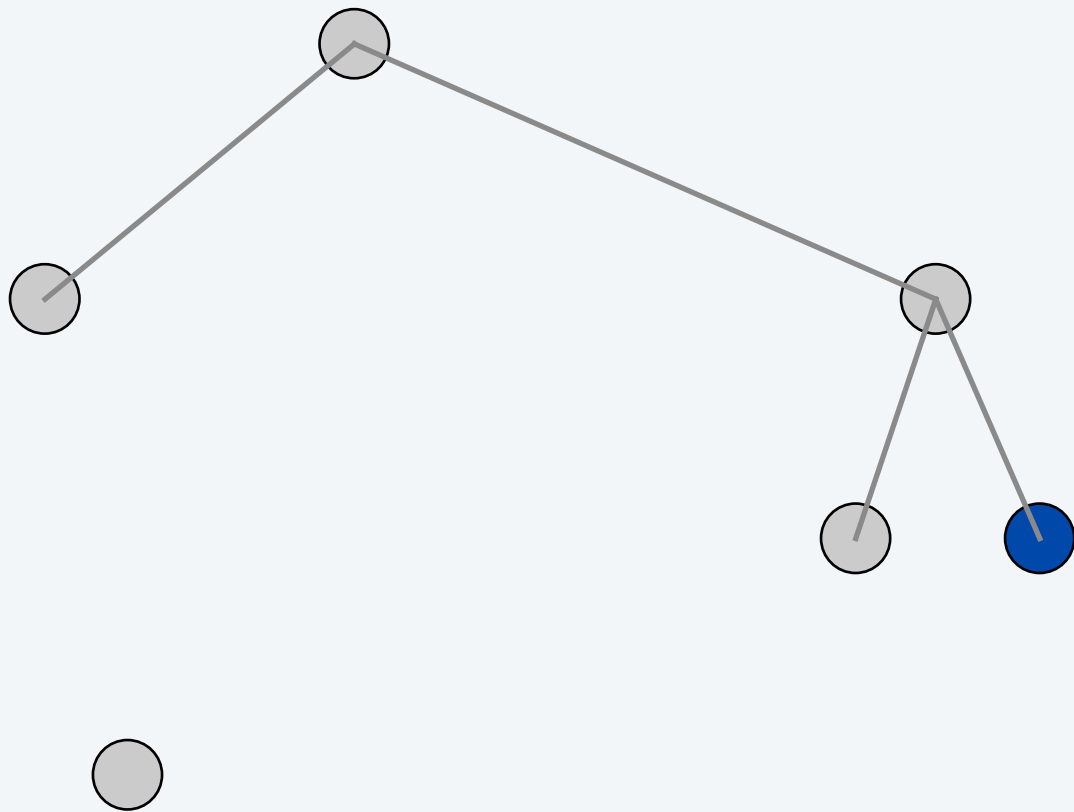
- Let (u, v) denote the incident edge.
- Add both v to the independent set.
- Delete u and v (and all incident edges).



Greedy independent set algorithm (for trees)

Repeatedly pick a leaf node v .

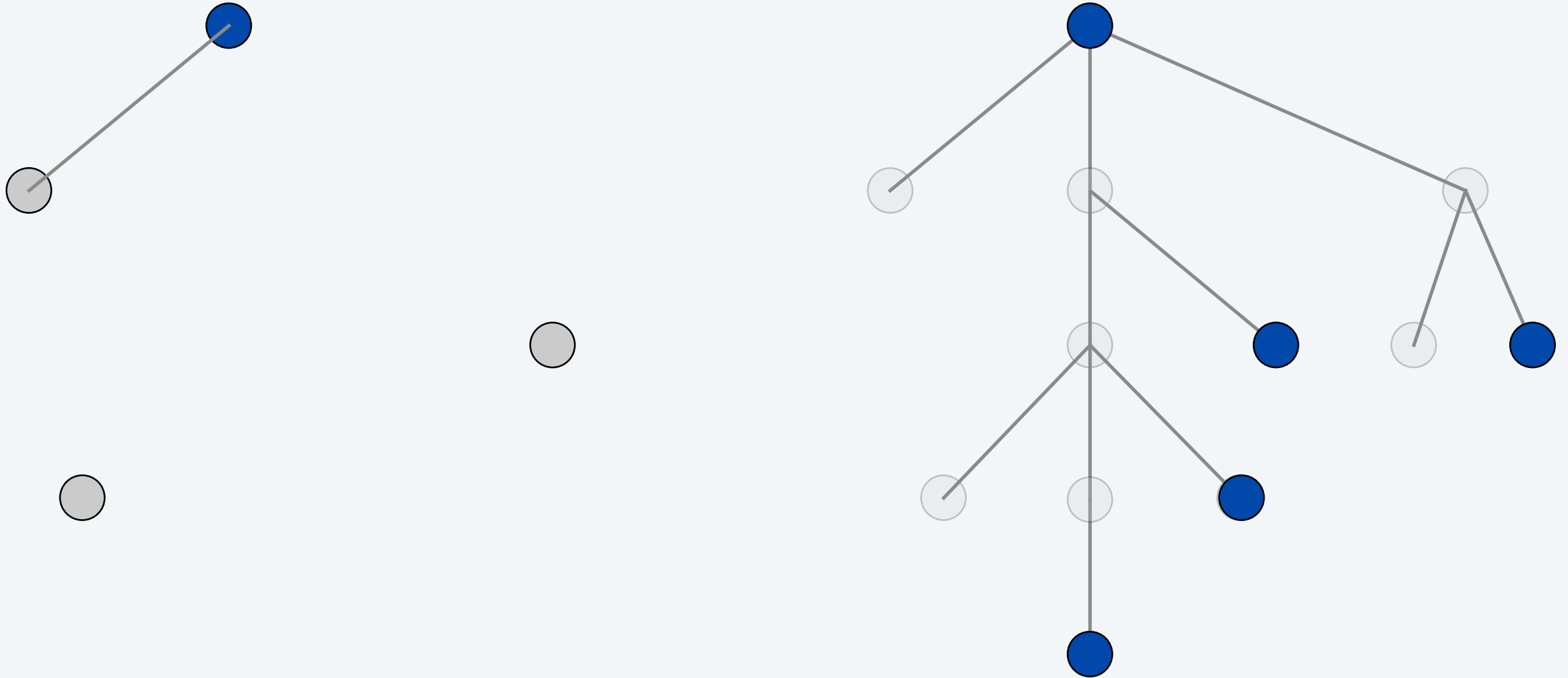
- Let (u, v) denote the incident edge.
- Add both v to the independent set.
- Delete u and v (and all incident edges).



Greedy independent set algorithm (for trees)

Repeatedly pick a leaf node v .

- Let (u, v) denote the incident edge.
- Add both v to the independent set.
- Delete u and v (and all incident edges).

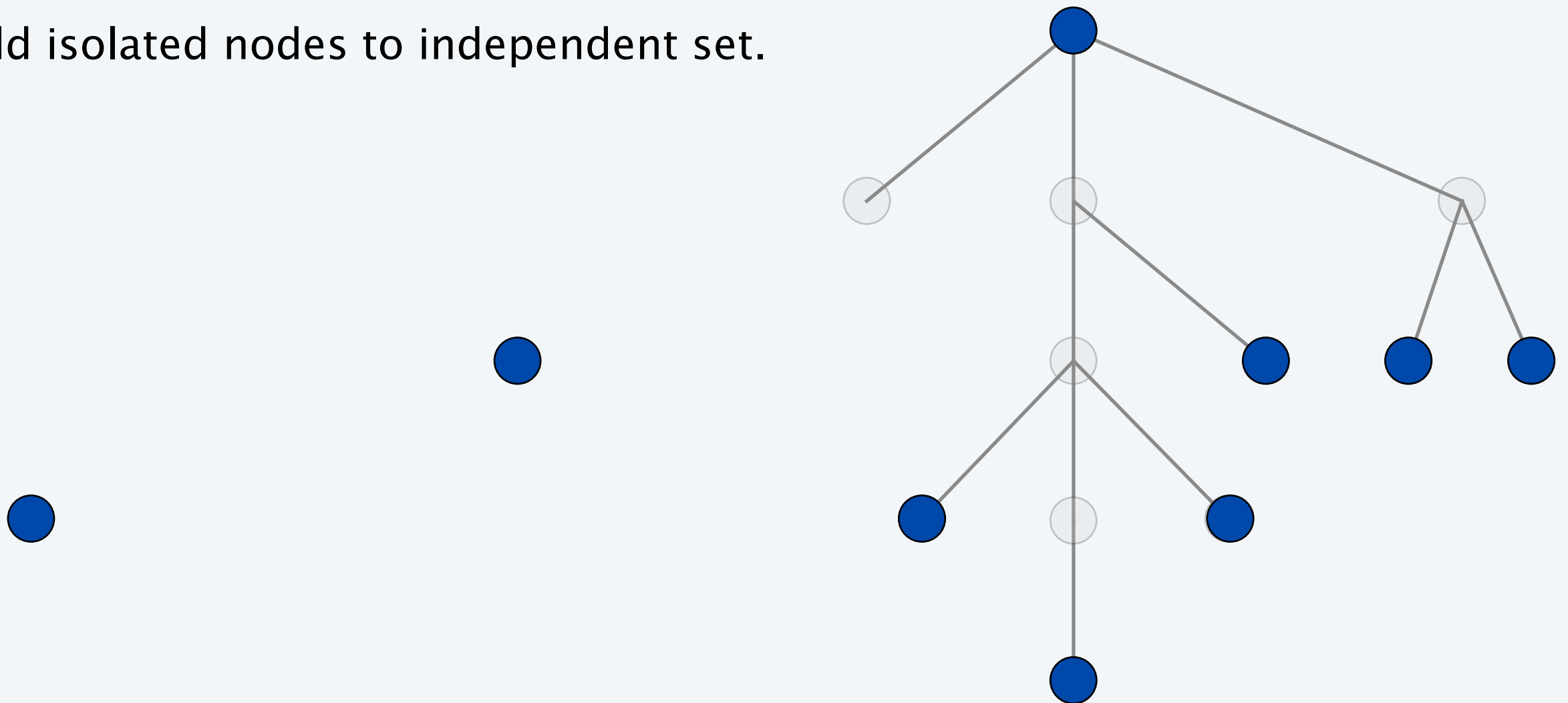


Greedy independent set algorithm (for trees)

Repeatedly pick a leaf node v .

- Let (u, v) denote the incident edge.
- Add both v to the independent set.
- Delete u and v (and all incident edges).

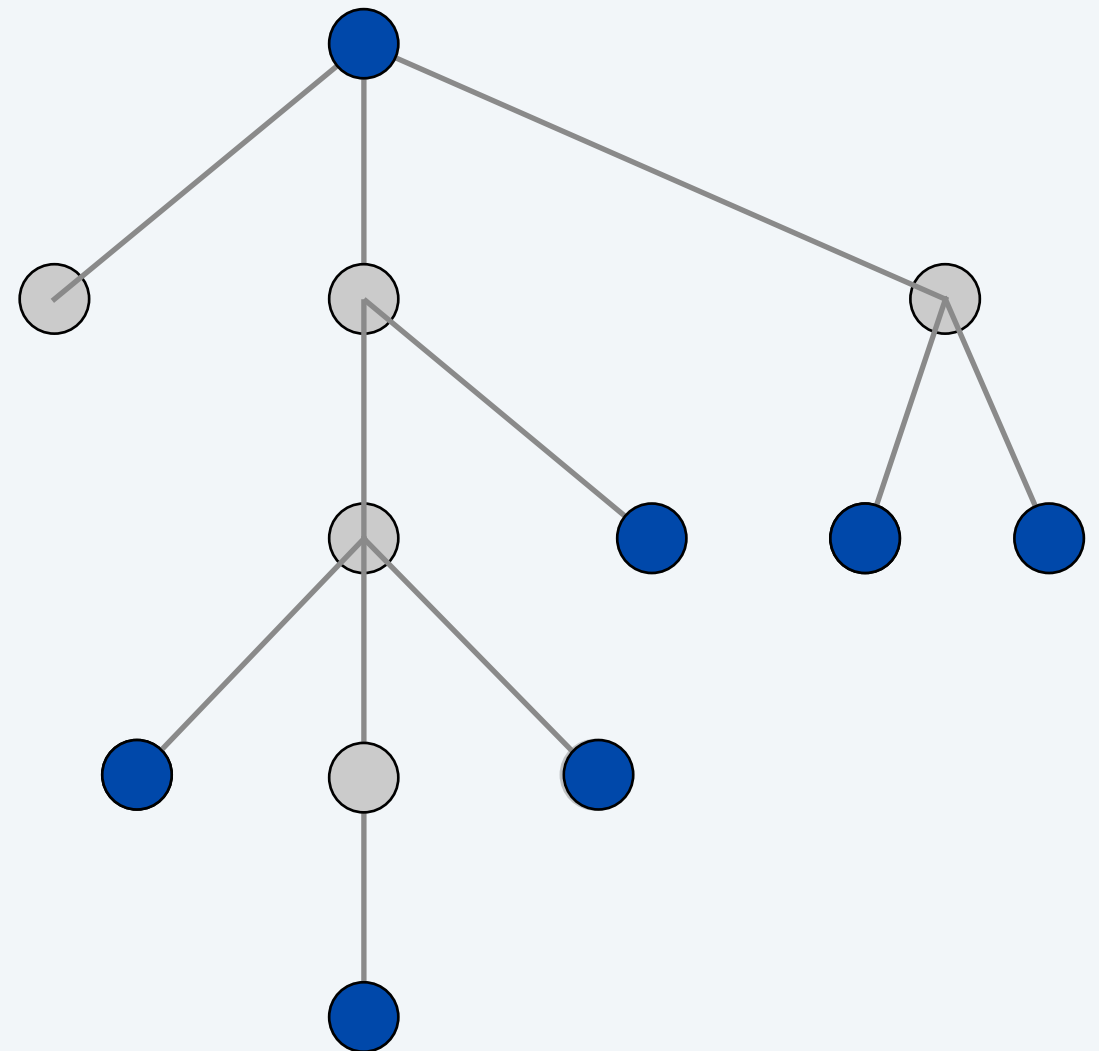
Add isolated nodes to independent set.



Greedy independent set algorithm (for trees)

Repeatedly pick a leaf node v .

- Let (u, v) denote the incident edge.
- Add both v to the independent set.
- Delete u and v (and all incident edges).



max-cardinality independent set